

Functions of Two Variables

Limits, Continuity and Partial Derivatives

Engineering Mathematics–I (CSE-1001)

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To the Student

This is the second note of **Module 2**, and it is where the subject genuinely changes character. Everything in Module 1 concerned a function of one variable — a point on a line, approached from the left or from the right. Here the input is a point in a *plane*, and a plane can be approached from infinitely many directions, along infinitely many curves.

Read the companion note *The Mean Value Theorems* first if you have not; and keep the Module 1 note on *Sequences* nearby, because the Sandwich Theorem does as much work here as it did there.

How to use this note:

1. Read the matching section after each lecture, working every example yourself before reading the solution.
2. Stop at every *Try it yourself* box — there are seven.
3. When a counterexample appears, do not skim it. In this topic the counterexamples *are* the theory: they are what force the definitions to take the shape they do.
4. Attempt the companion question bank once a section is finished.

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1 What Changes When There Are Two Variables

A function of one variable, $f : \mathbb{R} \rightarrow \mathbb{R}$, takes a number and returns a number. A function of two variables takes a *point of the plane*:

$$f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto f(x, y).$$

Its graph is the surface $z = f(x, y)$ sitting above the domain D .

Two variables is the normal case, not the exotic one

Almost nothing in engineering depends on one quantity alone. The temperature of a plate depends on *where* you measure it — two coordinates. The cost of manufacturing depends on labour and materials. The loss function of a neural network depends on every weight in the network, which is to say on millions of variables at once; and the training algorithm, gradient descent, is nothing but the two-variable theory of Section 5 scaled up. One-variable calculus is the special case. This note is where the general subject begins.

Here is the single structural difference that drives everything:

The approach problem

On the real line there are exactly **two** ways to approach a point a : from the left and from the right. That is why a one-variable limit exists precisely when two one-sided limits agree — a condition you can actually check.

In the plane there are **infinitely many** ways to approach a point (a, b) : along any straight line through it, along any parabola, along any spiral. For the limit to exist, *every single one* of these must give the same value. You cannot check them one at a time.

2 Limits of Functions of Two Variables

Definition 2.1 (Limit). Let f be defined on a disc around (a, b) , except possibly at (a, b) itself. We say

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

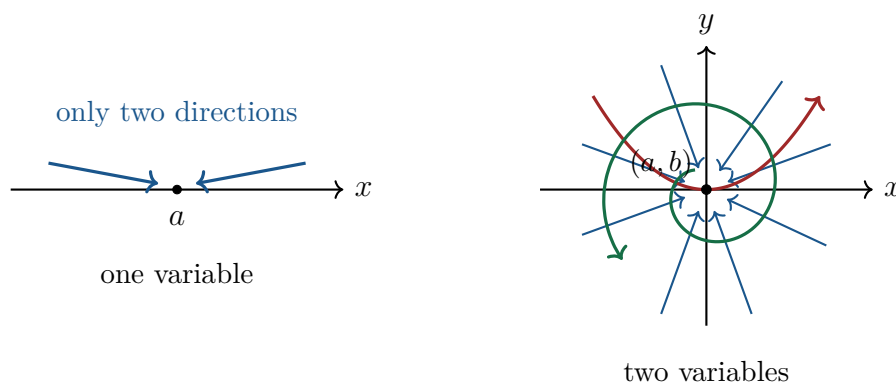


Figure 1: On the line, two approaches. In the plane, infinitely many — lines (blue), curves (red), spirals (green). The limit must agree along all of them.

if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \quad \implies \quad |f(x, y) - L| < \varepsilon.$$

In words: the values of f can be forced within ε of L by confining (x, y) to a small enough punctured disc about (a, b) — a *disc*, not an interval, and that is the whole difference.

2.1 Proving a limit does not exist: the path principle

Proposition 2.2 (Path principle). *If $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along one path, and $f(x, y) \rightarrow L_2$ along another path, with $L_1 \neq L_2$, then $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ **does not exist**.*

This follows immediately from the definition: a single L would have to serve both paths.

Example 2.3. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Solution. Approach along the x -axis, $y = 0$:

$$f(x, 0) = \frac{x^2 - 0}{x^2 + 0} = 1 \longrightarrow 1.$$

Approach along the y -axis, $x = 0$:

$$f(0, y) = \frac{0 - y^2}{0 + y^2} = -1 \longrightarrow -1.$$

Two paths, two different values. By Proposition 2.2 the limit does not exist. $\square$

2.2 The trap: paths can never prove a limit exists

Checking lines proves nothing

It is tempting to argue: “along $y = 0$ the limit is 0; along $x = 0$ it is 0; along every line $y = mx$ it is 0; therefore the limit is 0.”

This reasoning is invalid. The straight lines are only some of the paths. A function can agree along every line through the origin and still have no limit, because some *curve* disagrees. The next example is exactly such a function, and it is the reason this box exists.

The path principle is a **one-way tool**: it can prove a limit *does not* exist, never that it does.

Example 2.4. Investigate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$.

Solution. Along every straight line $y = mx$ (with $m \neq 0$):

$$f(x, mx) = \frac{x^2 \cdot mx}{x^4 + m^2 x^2} = \frac{mx^3}{x^2(x^2 + m^2)} = \frac{mx}{x^2 + m^2} \rightarrow \frac{0}{m^2} = 0.$$

Along the axes the function is identically 0. So *every line through the origin gives the limit 0*.

Now approach along the parabola $y = x^2$:

$$f(x, x^2) = \frac{x^2 \cdot x^2}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2} \quad \text{for every } x \neq 0,$$

so along this path the function is *constantly* $\frac{1}{2}$ and tends to $\frac{1}{2}$.

Two paths give 0 and $\frac{1}{2}$. The limit **does not exist** — even though every straight line said it did. $\square$

Why the parabola, and not a line?

The denominator contains x^4 and y^2 . These two terms are comparable in size exactly when y is comparable to x^2 — so that is the curve along which the numerator and denominator can balance. When you hunt for a bad path, look at the *exponents* and choose $y = kx^p$ with p matching the denominator’s own scaling.

2.3 Proving a limit does exist: squeeze and polar coordinates

Since paths cannot establish existence, we need a method that controls *all* approaches at once. Converting to polar coordinates does exactly that, because a single variable r then measures distance from the point regardless of direction.

Put $x = r \cos \theta$, $y = r \sin \theta$. Then $(x, y) \rightarrow (0, 0)$ is precisely $r \rightarrow 0^+$, and θ records the direction.

Example 2.5. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0$.

Solution. In polar coordinates,

$$f = \frac{(r \cos \theta)^2 (r \sin \theta)}{r^2} = r \cos^2 \theta \sin \theta.$$

Therefore

$$|f| = r |\cos^2 \theta \sin \theta| \leq r,$$

because $|\cos^2 \theta \sin \theta| \leq 1$. The bound r depends on r **alone** and not on θ , so it controls every direction simultaneously. As $r \rightarrow 0$ we get $|f| \rightarrow 0$, hence by the Sandwich Theorem

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0. \quad \square$$

The polar method has its own trap

The argument above works because the bound was r , *free of θ* . If you reach a bound such as r/θ or $\tan \theta$, which blows up for some directions, you have proved nothing: the convergence is not uniform in direction, and the limit may well fail to exist. Always check that the final bound involves r only.

Try it yourself

Evaluate, or show the limit does not exist:

$$(i) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} \quad (ii) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$$

Answers: (i) 0 — in polars it is $r \cos \theta \sin \theta$, bounded by r . (ii) does not exist — it is $\cos \theta \sin \theta$, with no r at all, so the value depends only on direction: 0 along $y = 0$ and $\frac{1}{2}$ along $y = x$.

3 Repeated (Iterated) Limits

There is a second, quite different thing one can compute: take the limits *one variable at a time*.

Definition 3.1 (Repeated limits). The two **repeated** (or **iterated**) limits at (a, b) are

$$\lim_{x \rightarrow a} \left(\lim_{y \rightarrow b} f(x, y) \right) \quad \text{and} \quad \lim_{y \rightarrow b} \left(\lim_{x \rightarrow a} f(x, y) \right).$$

These are not the same object as the **double limit** $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ of Section 2, in which both variables move together.

Example 3.2. For $f(x, y) = \frac{x - y}{x + y}$, compute both repeated limits at the origin.

Solution. Inner limit in y first:

$$\lim_{y \rightarrow 0} \frac{x - y}{x + y} = \frac{x}{x} = 1 \quad (x \neq 0), \quad \text{so} \quad \lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f \right) = 1.$$

Inner limit in x first:

$$\lim_{x \rightarrow 0} \frac{x - y}{x + y} = \frac{-y}{y} = -1 \quad (y \neq 0), \quad \text{so} \quad \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f \right) = -1.$$

The two repeated limits are 1 and -1 . $\square$

Proposition 3.3. *If the two repeated limits exist and are **different**, the double limit cannot exist.*

So in the example above, $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x+y}$ does not exist. This is a genuinely useful test, and often the quickest one available.

3.1 The relationship between the three limits — stated correctly

A false statement found in many texts

The following claim appears in a great many sets of notes:

“If the double limit exists, then both repeated limits exist and are equal to it.”

This is false. The existence of the double limit says nothing about whether the *inner* limit of a repeated limit exists at all.

Example 3.4 (The counterexample). Let

$$f(x, y) = \begin{cases} x \sin \frac{1}{y}, & y \neq 0, \\ 0, & y = 0. \end{cases}$$

Solution. **The double limit exists and is 0.** For all (x, y) ,

$$|f(x, y)| \leq |x| \leq \sqrt{x^2 + y^2},$$

so $f \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$ by the Sandwich Theorem.

But one repeated limit does not exist. Fix any $x \neq 0$ and let $y \rightarrow 0$. Then $\sin(1/y)$ oscillates through every value in $[-1, 1]$ without settling — taking the values $0, 1, 0, -1, \dots$ along $y = \frac{1}{2\pi}, \frac{1}{2.5\pi}, \frac{1}{3\pi}, \frac{1}{3.5\pi}, \dots$ — so $\lim_{y \rightarrow 0} f(x, y)$ **does not exist**. The repeated limit $\lim_{x \rightarrow 0} (\lim_{y \rightarrow 0} f)$ therefore does not exist either.

The double limit exists; a repeated limit does not. The claim in the box above is contradicted. $\square$

Here is the correct statement.

Theorem 3.5 (Double and repeated limits). *Suppose the double limit*

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

exists. If a repeated limit also exists, then it equals L. In particular, if all three exist, all three are equal.

The converse fails too, and for a different reason: the repeated limits can both exist and agree while the double limit does not.

Example 3.6. For $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ of Example 2.4, both repeated limits at the origin equal 0:

$$\lim_{y \rightarrow 0} f(x, y) = \frac{0}{x^4} = 0 \quad (x \neq 0), \quad \lim_{x \rightarrow 0} f(x, y) = \frac{0}{y^2} = 0 \quad (y \neq 0),$$

so both repeated limits are 0. Yet we showed the double limit does not exist, because the parabola $y = x^2$ gives $\frac{1}{2}$.

What you observe	What you may conclude	Why
Repeated limits differ	Double limit does not exist	Prop. 3.3
Repeated limits agree	Nothing	$\frac{x^2y}{x^4+y^2}$
Two paths disagree	Double limit does not exist	Prop. 2.2
All tested paths agree	Nothing	$\frac{x^2y}{x^4+y^2}$
Polar bound $\leq g(r) \rightarrow 0$	Double limit exists, $= L$	Sandwich

The asymmetry to remember

Every cheap test in this topic can only ever give a **negative** answer. To prove a limit *exists* you must bound the function in a way that is uniform over all directions — which in practice means the polar/squeeze method or the ε - δ definition itself.

4 Continuity

Definition 4.1 (Continuity at a point). f is **continuous at** (a, b) if

- (i) $f(a, b)$ is defined;
 - (ii) $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists;
 - (iii) the two are equal: $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$.
- f is continuous on a region if it is continuous at every point of it.

The structure is identical to the one-variable definition; only the meaning of “ $\rightarrow$ ” has changed.

Example 4.2. Is

$$f(x, y) = \begin{cases} \frac{x^2y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

continuous at the origin?

Solution. $f(0, 0) = 0$ is defined; we showed in Section 2 that the limit is 0; and the two agree. So **yes**, f is continuous at the origin. $\square$

Example 4.3. Is $g(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$ (with $g(0, 0) = 0$) continuous at the origin?

Solution. **No.** The limit does not exist (Section 2), so condition (ii) fails. No choice of value at the origin can repair this — the discontinuity is not removable. $\square$

	One variable	Two variables
Approaches to check	Two: left and right	Infinitely many directions and curves
Test for a limit	Compare two one-sided limits	No finite test; use squeeze or ε - δ
Disproving a limit	One-sided limits differ	Two paths differ
Typical difficulty	Routine	The hard part of the topic

5 Partial Derivatives

If a function depends on two variables, we can vary one and hold the other fixed. That is the whole idea.

Definition 5.1 (Partial derivatives). For $z = f(x, y)$,

$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad \frac{\partial f}{\partial y}(x, y) = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k},$$

provided the limits exist.

Notation. All of these mean the same thing:

$$\frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = f_x(x, y) = D_x f = z_x.$$

What a partial derivative measures

Slice the surface $z = f(x, y)$ with the vertical plane $y = \text{constant}$. The slice is an ordinary curve in one variable, and f_x is its ordinary slope. A partial derivative is a one-variable derivative taken along a slice — which is why every differentiation rule you already know still applies, with the other variable treated as a constant.

5.1 From first principles

Example 5.2. Find f_x and f_y from the definition for $f(x, y) = x^2y + y^3$.

Solution.

$$f_x = \lim_{h \rightarrow 0} \frac{[(x+h)^2y + y^3] - [x^2y + y^3]}{h} = \lim_{h \rightarrow 0} \frac{y(2xh + h^2)}{h} = \lim_{h \rightarrow 0} y(2x + h) = 2xy.$$

$$f_y = \lim_{k \rightarrow 0} \frac{[x^2(y+k) + (y+k)^3] - [x^2y + y^3]}{k} = \lim_{k \rightarrow 0} \frac{x^2k + 3y^2k + 3yk^2 + k^3}{k} = x^2 + 3y^2. \quad \square$$

In practice one differentiates directly, treating the other variable as a constant:

$$\frac{\partial}{\partial x} (x^2y + y^3) = 2xy,$$

since y^3 is constant as far as x is concerned.

5.2 Three or more variables

Nothing changes except the bookkeeping. For $w = f(x, y, z)$ we hold two variables fixed and vary the third.

Example 5.3. For $f(x, y, z) = x^2y + yz^3 + xz$, find all three first-order partials.

Solution.

$$f_x = 2xy + z, \quad f_y = x^2 + z^3, \quad f_z = 3yz^2 + x. \quad \square$$

Example 5.4. For $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ show that $f_x = x/f$, and write down f_y and f_z .

Solution. By the chain rule, $f_x = \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x}{f}$, and by symmetry $f_y = y/f$, $f_z = z/f$. $\square$

Try it yourself

Find all first-order partials of (i) $f = e^{xyz}$ and (ii) $f = \ln(x^2 + y^2 + z^2)$.

Answers: (i) $yz e^{xyz}$, $xz e^{xyz}$, $xy e^{xyz}$. (ii) $\frac{2x}{x^2 + y^2 + z^2}$, and similarly in y and z .

5.3 Higher-order partial derivatives

Differentiating again produces four second-order partials:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = (f_x)_y, \quad f_{yx} = (f_y)_x.$$

Fix your convention and keep it

The subscript notation is read **left to right in the order the differentiations are performed**: f_{xy} means “differentiate by x first, then by y ”. The operator notation runs the opposite way:

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x}.$$

Different books make different choices. Pick one — we use the one above throughout — and never mix the two inside a single calculation. Most errors in the next subsection are bookkeeping errors of exactly this kind.

Example 5.5. For $f(x, y, z) = \ln(xy + z^2)$, find f_y and then f_{yx} .

Solution.

$$f_y = \frac{x}{xy + z^2}.$$

Now differentiate with respect to x — and note that x appears in *both* the numerator and the denominator, so the quotient rule is needed in full:

$$f_{yx} = \frac{\partial}{\partial x} \left(\frac{x}{xy + z^2} \right) = \frac{(xy + z^2) \cdot 1 - x \cdot y}{(xy + z^2)^2} = \frac{z^2}{(xy + z^2)^2}.$$

$\square$

The slip this example is built around

A very common wrong answer here is

$$f_{yx} = \frac{(xy + z^2) \cdot 0 - x \cdot y}{(xy + z^2)^2} = \frac{-xy}{(xy + z^2)^2},$$

obtained by treating the numerator x as a constant while differentiating with respect to x . It is wrong in both sign and value. And note that it cannot be caught by the usual symmetry check — $-xy/(xy + z^2)^2$ is also symmetric in x and y , so it “passes” Clairaut’s test while being wrong. There is no substitute for doing the differentiation carefully.

5.4 Must the mixed partials be equal?

For every function you are likely to meet, $f_{xy} = f_{yx}$. But this is a theorem with hypotheses, not an identity — and it can fail.

Example 5.6 (Mixed partials that disagree). Let

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Show that $f_{xy}(0, 0) = -1$ while $f_{yx}(0, 0) = +1$.

Solution. Work from the definition at the origin, one step at a time.

Step 1 — f_x on the y -axis. For $y \neq 0$,

$$f_x(0, y) = \lim_{h \rightarrow 0} \frac{f(h, y) - f(0, y)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{hy(h^2 - y^2)}{h^2 + y^2} = \frac{y(-y^2)}{y^2} = -y,$$

and $f_x(0, 0) = 0$ as well, so $f_x(0, y) = -y$ for all y .

Step 2 — f_y on the x -axis. Symmetrically, for $x \neq 0$,

$$f_y(x, 0) = \lim_{k \rightarrow 0} \frac{1}{k} \cdot \frac{xk(x^2 - k^2)}{x^2 + k^2} = \frac{x(x^2)}{x^2} = x,$$

so $f_y(x, 0) = x$ for all x .

Step 3 — differentiate once more.

$$f_{xy}(0, 0) = \left. \frac{\partial}{\partial y} \right|_{y=0} f_x(0, y) = \left. \frac{\partial}{\partial y} \right|_{y=0} (-y) = -1,$$

$$f_{yx}(0, 0) = \left. \frac{\partial}{\partial x} \right|_{x=0} f_y(x, 0) = \left. \frac{\partial}{\partial x} \right|_{x=0} (x) = +1.$$

The two mixed partials differ. □

Which one is -1 ?

Under the convention $f_{xy} = (f_x)_y$ fixed above, the answer is $f_{xy}(0,0) = -1$ and $f_{yx}(0,0) = +1$. Under the opposite convention the labels swap. The numbers ± 1 are not in doubt; which label carries which sign depends entirely on the convention, so **state your convention before you start**. A solution that writes $f_{xy}(0,0) = +1$ without saying which order it means has not answered the question.

5.5 Clairaut's Theorem

What rules this behaviour out is continuity of the mixed partials.

Theorem 5.7 (Clairaut / Schwarz). *If f is defined on a disc containing (a,b) and both f_{xy} and f_{yx} are **continuous** on that disc, then*

$$f_{xy}(a,b) = f_{yx}(a,b).$$

Idea of the proof. Form the second difference

$$\Delta = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b).$$

Grouping the four terms one way and applying the one-variable Mean Value Theorem twice gives $\Delta = f_{xy}(p, q)hk$ for some point (p, q) in the rectangle; grouping them the other way gives $\Delta = f_{yx}(p', q')hk$ for another point. Dividing by hk and letting $(h, k) \rightarrow (0, 0)$, both interior points approach (a, b) , and continuity of the two mixed partials forces the two limits — hence $f_{xy}(a, b)$ and $f_{yx}(a, b)$ — to coincide. $\square$

Why Example 5.6 is not a contradiction

In that example f_{xy} and f_{yx} both exist everywhere, but they are **not continuous at the origin** — indeed they take different values there. The hypothesis of Theorem 5.7 fails, so the theorem has nothing to say, and no contradiction arises. This is exactly how a counterexample and a theorem are supposed to fit together.

6 A Warning: Partial Derivatives Can Exist Without Continuity

In one variable, differentiability at a point forces continuity there. In two variables the analogous statement with *partial* derivatives is false, and spectacularly so.

Example 6.1. Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Show that $f_x(0,0)$ and $f_y(0,0)$ both exist, yet f is not continuous at the origin.

Solution. **The partials exist.** Along the axes the function is identically zero: $f(h, 0) = 0$ and $f(0, k) = 0$. Hence

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0, \quad f_y(0,0) = \lim_{k \rightarrow 0} \frac{0 - 0}{k} = 0.$$

But f is not continuous there. Along $y = 0$, $f = 0$; along $y = x$,

$$f(x, x) = \frac{x^2}{2x^2} = \frac{1}{2} \quad (x \neq 0).$$

Two paths give 0 and $\frac{1}{2}$, so the limit does not exist and f is discontinuous at the origin. $\square$

Why this had to happen

$f_x(0, 0)$ and $f_y(0, 0)$ are computed from the behaviour of f along *two lines only* — the x -axis and the y -axis. The function is free to do anything it likes off those two lines, and this one does. Partial derivatives see two directions; continuity is about all of them.

The consequence for the whole module

“Both partial derivatives exist” is therefore a **much weaker** condition than one-variable differentiability. It is too weak to be the definition of differentiability in two variables, and the next note (*Differentiability, Chain Rule and Gradient*) builds the correct definition — one that *does* imply continuity. Everything that goes wrong in this section is the motivation for that definition.

7 Homogeneous Functions and Euler’s Theorem

One class of functions has partial derivatives tied together by a single identity, and it appears often enough to be worth a short section.

Definition 7.1 (Homogeneous function). $f(x, y)$ is **homogeneous of degree n** if

$$f(tx, ty) = t^n f(x, y) \quad \text{for all } t > 0.$$

Scaling both inputs by t scales the output by t^n . For example $f = x^3 + 3x^2y + y^3$ is homogeneous of degree 3; $f = \frac{x^3+y^3}{x+y}$ of degree 2; and $f = \tan^{-1}(y/x)$ of degree 0, since the ratio y/x is unchanged by scaling.

Theorem 7.2 (Euler’s Theorem). *If f is homogeneous of degree n and has continuous first partials, then*

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f.$$

Proof. Differentiate the identity $f(tx, ty) = t^n f(x, y)$ with respect to t , holding x and y fixed. By the chain rule the left side gives

$$x f_x(tx, ty) + y f_y(tx, ty),$$

and the right side gives $nt^{n-1}f(x, y)$. Now set $t = 1$:

$$x f_x(x, y) + y f_y(x, y) = n f(x, y). \quad \square$$

Example 7.3. Verify Euler’s theorem for $f = x^3 + 3x^2y + y^3$.

Solution. Here $n = 3$. We have $f_x = 3x^2 + 6xy$ and $f_y = 3x^2 + 3y^2$, so

$$x f_x + y f_y = 3x^3 + 6x^2y + 3x^2y + 3y^3 = 3(x^3 + 3x^2y + y^3) = 3f. \quad \square$$

Corollary 7.4 (Second-order form). *Under the same hypotheses, with continuous second partials,*

$$x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} = n(n-1) f.$$

Proof. Differentiate $xf_x + yf_y = nf$ partially with respect to x and multiply by x ; do the same with y ; add, and use $f_{xy} = f_{yx}$. $\square$

Try it yourself

Verify Euler's theorem for $f = \tan^{-1}\left(\frac{y}{x}\right)$.

Answer: f is homogeneous of degree 0, so the theorem predicts $xf_x + yf_y = 0$. Indeed $f_x = \frac{-y}{x^2+y^2}$ and $f_y = \frac{x}{x^2+y^2}$, and $xf_x + yf_y = \frac{-xy+xy}{x^2+y^2} = 0$.

8 Summary

The five things to carry forward

1. A limit in two variables must agree along **every** approach. Disagreement along two paths disproves existence; agreement along any number of paths proves nothing.
2. To prove a limit exists, bound $|f - L|$ by a quantity depending on r alone (polar coordinates), then squeeze.
3. Repeated limits are a different object from the double limit. Unequal repeated limits disprove the double limit; equal ones prove nothing; and the double limit can exist when a repeated limit does not.
4. A partial derivative is an ordinary derivative along a slice. Mixed partials agree when they are continuous (Clairaut), and can genuinely disagree when they are not.
5. **Existence of both partial derivatives does not imply continuity.** This is the central surprise of the note, and the reason the next note needs a stronger notion of differentiability.

Looking ahead. The next note asks what “differentiable” ought to mean for $f(x, y)$. The answer is not “both partials exist” — we have just seen that fail — but a condition saying the surface is well approximated by a *plane*. From it follow the chain rule, the gradient, tangent planes and linearisation.

Companion material: The Mean Value Theorems (Module 2, Note 1); Functions of Two Variables — Question Bank.